ROOM COMFORT MANAGEMENT SYSTEM USING ESP32

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Abstract

The Room Comfort Management System is an IoT-based solution designed to monitor indoor environmental conditions and automatically maintain a comfortable and healthy living environment. The system utilizes an ESP32 microcontroller along with temperature, humidity, light intensity, and soil moisture sensors to continuously collect environmental data. Based on predefined wellness criteria, the system analyzes room conditions, calculates a wellness score, generates recommendations, and automatically controls connected devices such as fans, lights, exhaust systems, and water pumps. This intelligent automation reduces human intervention while improving indoor comfort, energy efficiency, and plant health.

Introduction

Modern indoor environments require proper temperature, humidity, lighting, and air quality to ensure occupant comfort and productivity. Traditional monitoring systems only display sensor readings and rely on users to take corrective actions. This project extends beyond monitoring by incorporating automated decision-making and device control.

The Smart Indoor Wellness Management System continuously evaluates indoor conditions and takes appropriate actions to maintain optimal environmental parameters. By integrating sensing, analysis, and automation, the system creates a smart indoor ecosystem capable of responding intelligently to changing conditions.

Objectives

1. To monitor indoor environmental conditions in real time.
2. To measure temperature, humidity, light intensity, and soil moisture.
3. To calculate a wellness score based on environmental conditions.

4. To provide recommendations for improving comfort.
5. To automate electrical devices based on sensor readings.
6. To reduce manual intervention through intelligent control.

Components Required

Hardware Components

Component	Purpose
ESP32	Main Controller
DHT11/DHT22	Temperature and Humidity Measurement
LDR Sensor	Light Intensity Detection
Soil Moisture Sensor	Plant Monitoring
Relay Module	Switching Electrical Devices
DC Fan	Temperature Control
LED Lamp	Smart Lighting
Mini Water Pump	Automatic Plant Watering
Breadboard & Jumper Wires	Circuit Connections

Software Requirements

- Arduino IDE
- ESP32 Board Package
- Blynk/IoT Dashboard (Optional)
- Google Stitch (Optional UI Design)

System Architecture

Temperature Sensor

Humidity Sensor

Light Sensor

Soil Moisture Sensor

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ESP32 Controller

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Data Processing & Wellness Analysis

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Decision Engine

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Relay Control System

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Fan / Light / Exhaust Fan / Water Pump

Methodology

The ESP32 continuously reads data from all sensors. The collected values are compared against predefined thresholds to evaluate environmental comfort.

The system performs two primary functions:

Monitoring

- Displays sensor readings
- Calculates wellness score
- Identifies environmental issues
- Generates recommendations

Automation

- Activates devices automatically when unfavorable conditions are detected
- Maintains optimal room conditions without user intervention

Working Principle

Temperature Control

Condition:
Temperature > 32°C

Action:
Fan ON

Output:
"High temperature detected. Cooling system activated."

Lighting Control

Condition:
Light Intensity Below Threshold

Action:
LED Light ON

Output:
"Low lighting detected. Room lighting activated."

Humidity Control

Condition:
Humidity > 80%

Action:
Exhaust Fan ON

Output:
"High humidity detected. Ventilation activated."

Plant Care Automation

Condition:
Soil Moisture Below Threshold

Action:
Water Pump ON

Output:
"Plant watering initiated."

Wellness Score Calculation

The system begins with a score of 100 and deducts points based on unfavorable environmental conditions.

Condition	Penalty
High Temperature	-20
Low Humidity	-15
High Humidity	-15
Low Light	-20
Dry Soil	-10

Example

Initial Wellness Score = 100

High Temperature = -20

Low Light = -20

Dry Soil = -10

Final Wellness Score = 50%

Status: Moderate Comfort

Sample Output

Temperature: 35°C

Humidity: 45%

Light Intensity: Low

Soil Moisture: Dry

Wellness Score: 50%

Issues Detected:

- High Temperature
- Poor Lighting
- Plant Requires Watering

Actions Taken:

- Fan Activated
- Light Activated
- Water Pump Activated

Advantages

- Real-time environmental monitoring
- Automatic device control
- Improved indoor comfort
- Reduced energy wastage
- Low-cost implementation
- User-friendly operation
- Scalable IoT architecture

Applications

- Smart Homes
- Hostels
- Offices
- Libraries

- Classrooms
- Indoor Gardening Systems
- Smart Building Automation

Future Scope

1. Mobile App Integration
2. Cloud Data Storage and Analytics
3. Voice Assistant Support
4. Machine Learning-Based Prediction
5. Weather-Aware Automation
6. Energy Optimization Algorithms
7. Smart Air Quality Monitoring

Conclusion

The Smart Indoor Wellness Management System successfully integrates environmental monitoring, intelligent analysis, and automation into a single IoT platform. By combining multiple sensors with automated control mechanisms, the system creates a comfortable, healthy, and energy-efficient indoor environment. The project demonstrates how IoT technology can be utilized to enhance quality of life through intelligent environmental management and automated decision-making.